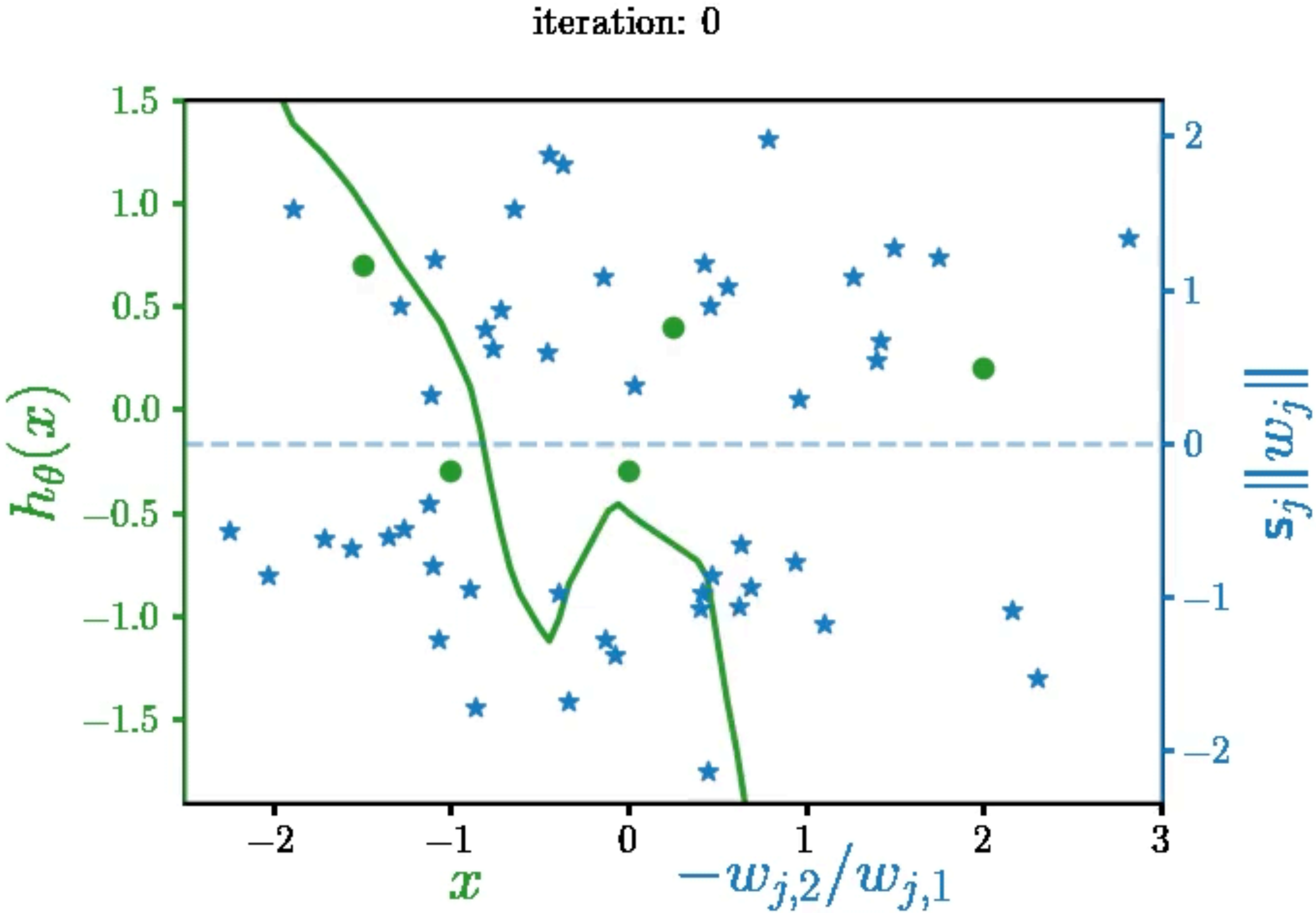
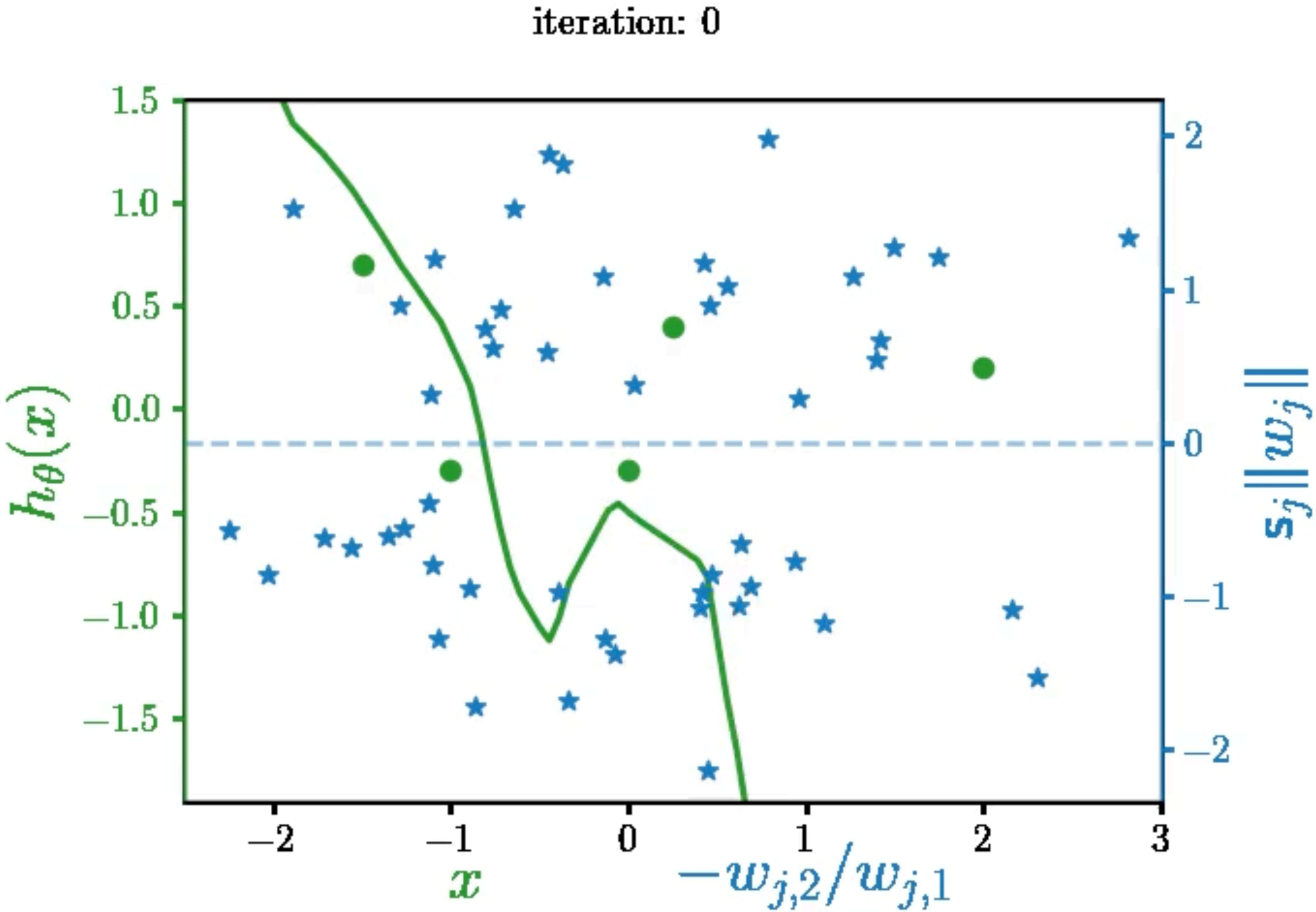
Illustration of lazy training

12


iteration: 0
1.5
1.0
0.5
0.0
−0.5
−1.0
−1.5
$h_\theta(x)$
2
1
0
−1
−2
$\|w_j\|$
−2
−1
0
1
2
3
x
$-w_{j,2}/w_{j,1}$

Data $x_i \in \mathbb{R}^2$
(bias = second coordinate)

- Dot: data points
- Line: estimated function
- Star: single neuron (w_j, a_j)
  x value $=$ kink position
  y value $= a_j$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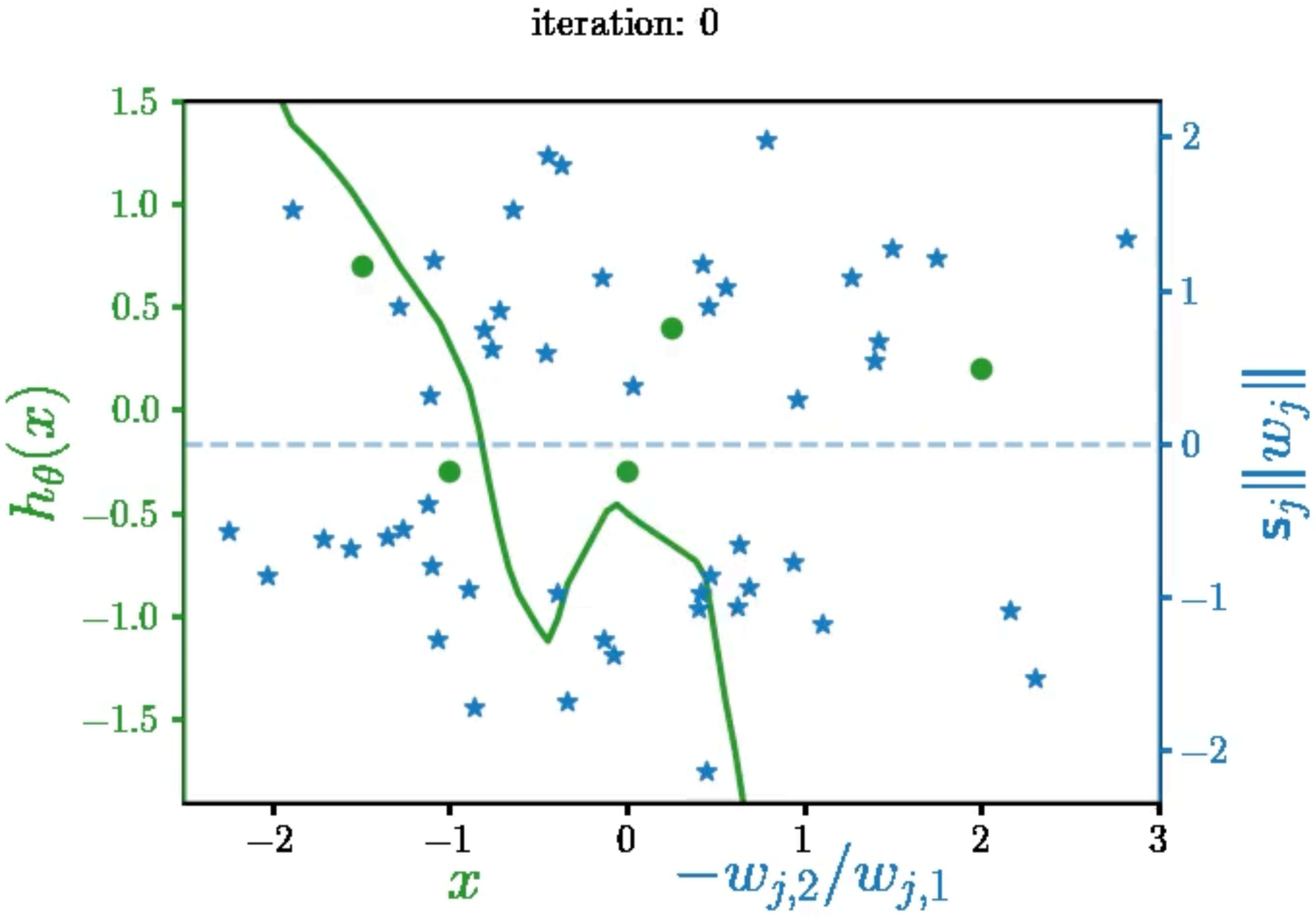

Rich regime

scale of initialisation

Rich regime: $(w_i^0, a_i^0) \sim \frac{\lambda}{\sqrt{m}} D$ → feature learning happens

In the following $\lambda \ll 1$

D such that $\begin{cases} |a_i| \geq \|w_i\| & \text{for any } i \\ \sum_{i=1}^m (a_i^0)^2 \leq \lambda \end{cases}$

→ balanced initialisation

→ finite variance

Example: $\begin{cases} w_i \sim \lambda_m \mathcal{U}(B(0,1)) \\ a_i = \pm \lambda_m \end{cases}$

Lemma (balancedness)

$$\boxed{(a_j^t)^2 - \|w_j^t\|^2 = \text{constant}}$$

a_i^t **has a constant sign**